

06 Jul 2026 11:37:18 ET | 25 pages

Taiwan Semiconductors

Mediatek and GUC riding on Google; Alchip more back-end loaded and capacity allocation is key

CITI'S TAKE

Investors have been optimistic about AI ASIC surpassing GPU in AI computing since last year, yet due to a more fragmented supply chain and tight component/material supply, such as wafer foundry, advanced packaging, HBM and ABF, we expect Taiwanese design service companies with better supply chain management capability to stand out with positive growth outlooks into 2027. We expect Mediatek and GUC to see the strongest AI ASIC growth momentum into next year. We prefer Mediatek most thanks to its higher value content upside with its increasing contribution from I/O, top die and potentially optics and memory custom based die in the longer term.

Mediatek - AI ASIC revenue to rise 50% in 2027, continue doubling into 2028 — We estimate Mediatek's AI ASIC revenue to reach US\$18bn in 2027 and expect it to further rise to US\$40bn in 2028 (from previous expectations of US\$13.6bn/18bn for 2027/28) mainly due to higher dollar content increase thanks to its value added not just in I/O die but also compute die. From 2028, we see Mediatek will contribute more on multiple chip tile of its client's chiplet design. Mediatek is also working on multiple opportunities in AI ASIC projects – not only chip design but also hardware L6 architecture. We expect its AI revenue to exceed its current existing business in 2 years.

GUC - CPU business upside intact in next 2 years; automotive business also showing good potential in 2028 — Thanks to solid CPU ASIC demand from its US CSP clients, we expect GUC's growth into 2027/28 to be stronger with more supply chain support. Its robotic/automotive for US client will also start contributing in late 2027/early 2028. Although its customers may look for another foundry's support besides TSMC, we expect GUC's solid partnership with TSMC to continue to be the main source for the automotive/robotic project. And GUC's CPU projects' visibility is intact through 2028. The company is also working on consumer electronics for CSP client. Along with its interconnect IP and CPO technology, we believe GUC's ASIC business growth trend remains intact.

Alchip - new product to kick off in 3Q26, suggest revisiting once 2026 expectations reset; 2027 outlook intact — We expect Alchip's revenue to snap back substantially in 2H26 thanks to its client's N3 AI accelerator ramp-up. However, due to potential capacity constraints, its client's COT business model and diversification strategy, Alchip's near-term momentum could be capped. Yet GM may see upside thanks to Alchip's back-end design contribution for more complex chip design. We would expect stronger 2027 growth once Alchip can secure more order allocation and supply.

Mediatek remains our favorite stock; Our pecking order: Mediatek > GUC > Alchip — We prefer Mediatek in AI ASIC design service thanks to its better front-end design capability in advanced node, interconnect, advanced packaging and upstream semiconductor supply chain. We also see GUC's diversification in AI ASIC/automotive/robotic, consumer product projects and strong CPU upside to support growth in the next 2-3 years. Alchip's momentum would be more back-end loaded, and we expect to see better growth into next year.

Laura (Chia Yi) Chen^{AC}

+886-2-8726-9090

laura.cy.chen@citi.com

Jack Chen

+886-2-8726-9091

jack1.chen@citi.com

Nicholas Lai

+886-2-8726-9093

nicholas.lai@citi.com

See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

Data Summary

															Current Fiscal Year		Next Fiscal Year	
Company	Ticker	Ccy	Price	Mkt Cap (M)	Date & Time	Rating		Short-Term View	Target Price		ESPR (%)	Div Yld (%)	ETR (%)	Last Rpt Yr	EPS		EPS	
						Old	New		Old	New					Old	New	Old	New
Alchip	3661.TW	NT\$	4,215.00	346,493	06 Jul 13:30	1	nc	Upside	6,200.00	nc	47.1	0.8	47.9	Dec-25	90.64	103.06	155.14	168.72
GUC	3443.TW	NT\$	4,605.00	617,125	06 Jul 13:30	1H	nc	Upside	5,550.00	6,000.00	30.3	0.4	30.7	Dec-25	51.08	49.23	88.08	101.56
MediaTek	2454.TW	NT\$	4,125.00	6,616,069	06 Jul 13:30	1	nc	Upside^	5,950.00	6,055.00	46.8	1.3	48.1	Dec-25	68.28	68.11	124.07	119.47
1 = Buy, 2 = Neutral, 3 = Sell, H = High Risk						ESPR = Expected Share Price Return, ETR = Expected Total Return, nc = no change												
Source: Citi Research						^Catalyst Watch												

Earnings Estimates

Company Name	Ticker	Last Rpt Year	Currency	Last Reported Year					Current Fiscal Year					Next Fiscal Year				
				1Q	2Q	3Q	4Q	FY0	1Q	2Q	3Q	4Q	FY1	1Q	2Q	3Q	4Q	FY2
Alchip	3661.TW	Dec-25	NT\$	17.69	16.02	16.04	17.92	67.68	17.25	24.64	25.57	35.61	103.06	31.34	37.86	45.62	53.90	168.72
Old		Dec-25	NT\$	17.69	16.02	16.04	17.92	67.68	17.25	20.78	25.19	27.42	90.64	28.29	33.90	44.35	48.61	155.14
GUC	3443.TW	Dec-25	NT\$	7.14	5.81	6.44	8.61	28.01	10.38	11.59	12.94	14.32	49.23	16.00	22.13	27.81	35.63	101.56
Old		Dec-25	NT\$	7.14	5.81	6.44	8.61	28.01	12.23	11.59	12.94	14.32	51.08	15.02	19.58	24.75	28.74	88.08
MediaTek	2454.TW	Dec-25	NT\$	18.43	17.50	15.84	14.40	66.17	15.17	15.94	18.26	18.75	68.11	21.88	24.98	32.91	39.70	119.47
Old		Dec-25	NT\$	18.43	17.50	15.84	14.40	66.17	15.17	15.94	18.26	18.91	68.28	21.67	24.66	34.04	43.69	124.07

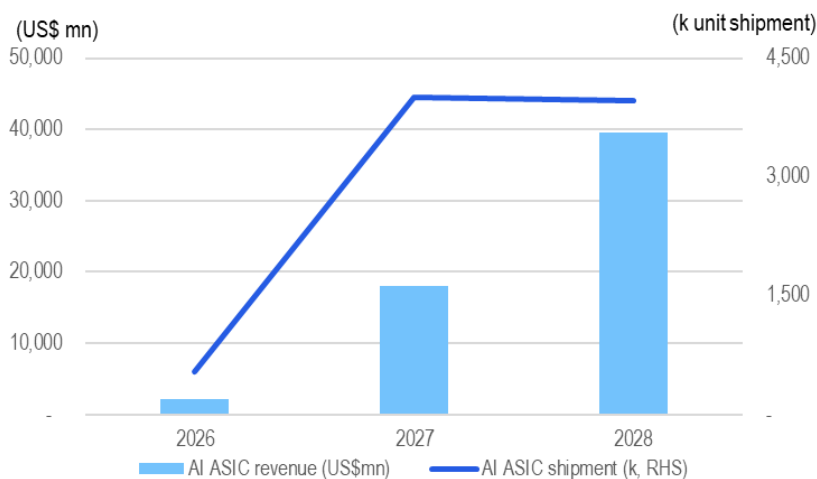
Source: Citi Research

Mediatek- AI ASIC revenue to rise 50% in 2027, continue doubling into 2028

We expect Mediatek to gain further market share in Google's growing TPU opportunity thanks to its better supply chain management capability and advanced node design capability. Other than N3 top die, Mediatek has been working on global leading 300G Serdes I/O for its next generation AI accelerator. Mediatek is also working on CPO technology with various partners such as Arya Labs and Microsoft in Micro LED CPO light source. With large-scale TPU revenue contribution, we now expect Mediatek's TPU shipment to reach 500k/4m/4m in 2026/2027/2028 with revenue contribution at US\$2bn/US\$18bn/US\$40bn.

Mediatek - Maintain Buy and raise TP to NT\$6,055 — In the near term, despite smartphone demand remaining lukewarm, thanks to better ASP we expect Mediatek to beat its 2Q revenue guidance and see c5% sequential growth in 3Q26. We slightly lower our 2026/2027 earnings projections mainly on smartphone weakness yet lift our 2028E earnings by 9%. We also raise our target price from NT\$5,950 to NT\$6,055 (35x PER to 2027/2028 EPS, which is 1.5-std above its 3-year average forward P/E given the promising outlook from its AI ASIC business starting from 2H26).

Figure 1. Mediatek – AI ASIC revenues and shipment forecast



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

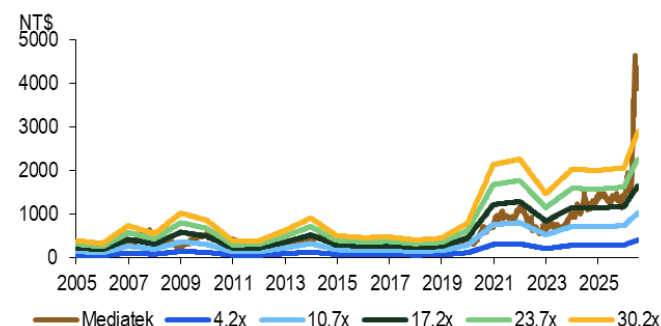
Figure 2. Mediatek – Estimates Revisions

(NT\$m)	2026E			2027E			2028E		
	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	616,651	619,460	-0.5%	1,083,075	1,128,745	-4.0%	2,110,784	1,928,342	9.5%
YoY growth	3%	4%		76%	82%		95%	71%	
Gross profit	284,934	286,058	-0.4%	492,443	513,381	-4.1%	943,367	861,801	9.5%
Opex	185,062	185,891	-0.4%	304,633	317,447	-4.0%	568,994	519,906	9.4%
Operating profit	99,872	100,167	-0.3%	187,810	195,934	-4.1%	374,373	341,894	9.5%
Pre-tax profit	121,102	121,397	-0.2%	211,699	219,823	-3.7%	399,316	366,837	8.9%
Net income	108,435	108,701	-0.2%	190,184	197,514	-3.7%	359,452	330,149	8.9%
EPS (NT\$)	68.11	68.28	-0.2%	119.47	124.07	-3.7%	225.79	207.39	8.9%
Gross margin	46.2%	46.2%	+0.0 ppt	45.5%	45.5%	-0.0 ppt	44.7%	44.7%	+0.0 ppt
Opex ratio	30.0%	30.0%	+0.0 ppt	28.1%	28.1%	+0.0 ppt	27.0%	27.0%	-0.0 ppt
Operating margin	16.2%	16.2%	+0.0 ppt	17.3%	17.4%	-0.0 ppt	17.7%	17.7%	+0.0 ppt
Net margin	17.6%	17.5%	+0.0 ppt	17.6%	17.5%	+0.1 ppt	17.0%	17.1%	-0.1 ppt

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

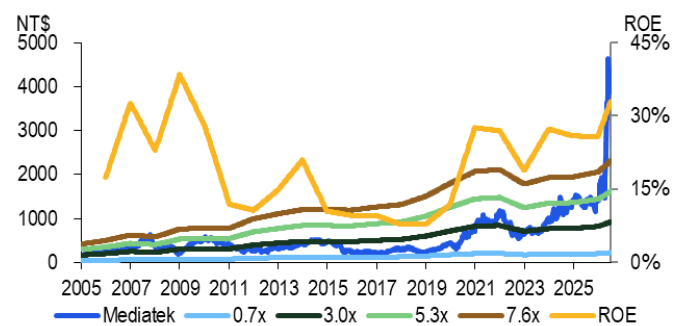
Figure 3. Mediatek – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 4. Mediatek – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 5. Mediatek – Forecast Summary

NT\$m	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2025	2026E	2027E	2028E
Net sales	150,369	142,097	150,188	149,151	143,591	150,023	173,885	203,214	236,298	287,482	356,081	595,966	616,651	1,083,075	2,110,784
Gross profit	73,878	66,112	69,281	69,055	66,644	70,488	78,747	90,891	104,532	132,787	164,233	283,080	284,934	492,443	943,367
OPEX	44,499	43,924	47,431	46,164	43,795	43,807	51,296	57,916	66,163	80,495	100,059	179,610	185,062	304,633	568,994
Operating profit	29,379	22,188	21,850	22,891	22,848	26,681	27,451	32,975	38,369	52,292	64,174	103,470	99,872	187,810	374,373
Total Non-OP	3,849	7,772	5,298	4,128	5,497	5,761	5,844	5,854	5,930	6,008	6,096	21,418	21,230	23,889	24,943
Pre-tax profit	33,228	29,960	27,147	27,019	28,345	32,442	33,295	38,829	44,299	58,301	70,270	124,888	121,102	211,699	399,316
Income tax	5,163	4,509	4,074	2,643	2,773	3,173	3,257	3,798	4,333	5,703	6,874	18,770	11,846	20,708	39,060
Net profit	27,848	25,221	22,925	24,154	25,369	29,067	29,844	34,826	39,764	52,397	63,197	105,319	108,435	190,184	359,452
EPS (NT\$)	17.50	15.84	14.40	15.17	15.94	18.26	18.75	21.88	24.98	32.91	39.70	66.17	68.11	119.47	225.79
Margins (%)															
Gross profit	49.1	46.5	46.1	46.3	46.4	47.0	45.3	44.7	44.2	46.2	46.1	47.5	46.2	45.5	44.7
OPEX to Sales Ratio	29.6	30.9	31.6	31.0	30.5	29.2	29.5	28.5	28.0	28.0	28.1	30.1	30.0	28.1	27.0
Operating profit	19.5	15.6	14.5	15.3	15.9	17.8	15.8	16.2	16.2	18.2	18.0	17.4	16.2	17.3	17.7
Pre-tax profit	22.1	21.1	18.1	18.1	19.7	21.6	19.1	19.1	18.7	20.3	19.7	21.0	19.6	19.5	18.9
Net profit	18.5	17.7	15.3	16.2	17.7	19.4	17.2	17.1	16.8	18.2	17.7	17.7	17.6	17.6	17.0
Y/Y(%)															
Net sales	18.1	7.8	8.8	(2.7)	(4.5)	5.6	15.8	36.2	64.6	91.6	104.8	12.3	3.5	75.6	94.9
Gross profit	18.9	2.7	3.4	(6.4)	(9.8)	6.6	13.7	31.6	56.9	88.4	108.6	7.5	0.7	72.8	91.6
OPEX	19.7	8.5	4.0	5.5	(1.6)	(0.3)	8.1	25.5	51.1	83.7	95.1	11.6	3.0	64.6	86.8
Operating profit	17.7	(7.0)	2.0	(23.8)	(22.2)	20.3	25.6	44.1	67.9	96.0	133.8	1.0	(3.5)	88.1	99.3
Pre-tax profit	13.8	5.3	3.6	(21.8)	(14.7)	8.3	22.6	43.7	56.3	79.7	111.1	4.5	(3.0)	74.8	88.6
Net profit	8.3	(0.5)	(3.6)	(17.6)	(8.9)	15.3	30.2	44.2	56.7	80.3	111.8	(1.0)	3.0	75.4	89.0
Q/Q(%)															
Net sales	(1.9)	(5.5)	5.7	(0.7)	(3.7)	4.5	15.9	16.9	16.3	21.7	23.9				
Gross profit	0.1	(10.5)	4.8	(0.3)	(3.5)	5.8	11.7	15.4	15.0	27.0	23.7				
Operating profit	(2.2)	(24.5)	(1.5)	4.8	(0.2)	16.8	2.9	20.1	16.4	36.3	22.7				
Pre-tax profit	(3.8)	(9.8)	(9.4)	(0.5)	4.9	14.5	2.6	16.6	14.1	31.6	20.5				
Net profit	(5.0)	(9.4)	(9.1)	5.4	5.0	14.6	2.7	16.7	14.2	31.8	20.6				

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

GUC- Solid order visibility into 2028

We note that there is rising concern about GUC's CPU ASIC order sustainability and potential foundry strategy diversification from its major clients. In our view, although its customers may look for more foundry sources' support, TSMC would remain the main source; thus, GUC's order visibility in the next 2-3 years remains intact. More importantly, GUC has developed a comprehensive turnkey model that spans specification, physical implementation, advanced packaging, manufacturing enablement, and production support. GUC has emerged as a provider of chiplet-related IP. Over the past two years, GUC has introduced HBM3 and HBM4 controller and PHY solutions, UCIe chiplet interfaces, and complete 2.5D/3D ASIC services built around CoWoS and SolC. We would expect stronger earnings momentum over the next two years thanks to strong ASIC projects pipeline and better capacity support from TSMC.

GUC - Maintain Buy/1H and raise TP to NT\$6,000 — We estimate GUC's CPUs revenue from two US CSPs would be over US\$1.5bn in 2027 with revenue contribution to be over 50%. Given strong order outlook from its key ASIC project, we raise our 2027/2028 earnings estimates by 15%/18%. We also maintain our Buy/High Risk rating on GUC and lift TP from NT\$5,550 to NT\$6,000, based on 45x PER to the average of our 2027/28E EPS, slightly above its historical 3-year PER average, which we believe is justified by its strong earnings outlook supported by solid demand from its AI projects with two US leading CSPs and growth upside from its US EV clients.

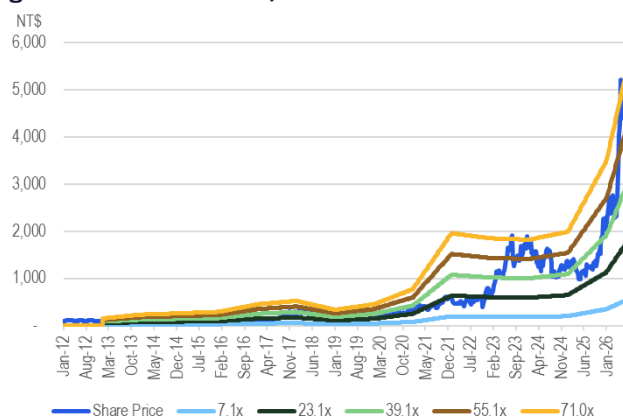
Figure 6. GUC – Estimates Revisions

(NT\$m)	2026E			2027E			2028E		
	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	55,985	55,985	0%	99,597	92,193	8%	160,800	149,733	7%
YoY growth	64%	64%		78%	65%		61%	62%	
Gross profit	13,002	13,293	-2%	23,885	21,159	13%	37,959	33,157	14%
Opex	5,561	5,561	0%	8,171	7,567	8%	13,217	12,305	7%
Operating profit	7,441	7,732	-4%	15,714	13,592	16%	24,742	20,851	19%
Pre-tax profit	7,753	8,044	-4%	15,994	13,872	15%	25,022	21,131	18%
Net income	6,629	6,878	-4%	13,675	11,860	15%	21,394	18,067	18%
EPS (NT\$)	49.23	51.08	-4%	101.56	88.08	15%	158.89	134.18	18%
Gross margin	23.2%	23.7%	-0.5 ppt	24.0%	23.0%	+1.0 ppt	23.6%	22.1%	+1.5 ppt
Opex ratio	9.9%	9.9%	+0.0 ppt	8.2%	8.2%	-0.0 ppt	8.2%	8.2%	+0.0 ppt
Operating margin	13.3%	13.8%	-0.5 ppt	15.8%	14.7%	+1.0 ppt	15.4%	13.9%	+1.5 ppt
Net margin	11.8%	12.3%	-0.4 ppt	13.7%	12.9%	+0.9 ppt	13.3%	12.1%	+1.2 ppt

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

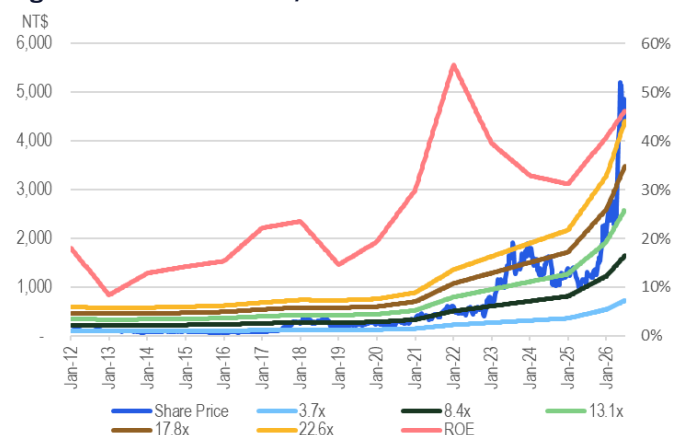
Figure 7. GUC – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 8. GUC – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 9. GUC – Forecast Summary

GUC	2025				2026				2027				2025	2026E	2027E	2028E
(NT\$ in Mn, year-end Dec)	1Q	2Q	3Q	4Q	1Q	2QE	3QE	4QE	1QE	2QE	3QE	4QE				
Revenue	7,024	6,105	8,613	12,400	11,448	13,032	14,889	16,616	18,271	21,910	27,856	31,560	34,141	55,985	99,597	160,800
COGS	-5,051	-4,071	-6,527	-10,038	-8,626	-9,934	-11,536	-12,887	-14,176	-16,655	-21,291	-23,590	-25,687	-42,983	-75,712	-122,841
Depreciation costs	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-6	-23	-22	-22	-22
Gross Profit	1,973	2,034	2,086	2,361	2,822	3,098	3,353	3,729	4,095	5,255	6,566	7,969	8,454	13,002	23,885	37,959
Operating Expense	-878	-1,090	-1,031	-1,105	-1,289	-1,342	-1,385	-1,545	-1,644	-1,840	-2,256	-2,430	-4,104	-5,561	-8,171	-13,217
SG&A expenses	-224	-253	-218	-224	-286	-300	-342	-382	-365	-372	-474	-537	-920	-1,311	-1,748	-2,826
R&D expenses	-800	-837	-813	-880	-1,003	-1,043	-1,042	-1,163	-1,279	-1,468	-1,783	-1,894	-3,330	-4,251	-6,423	-10,391
EBIT	1,095	944	1,055	1,257	1,533	1,756	1,968	2,184	2,451	3,414	4,309	5,539	4,350	7,441	15,714	24,742
Pre-Tax Profit	1,143	929	1,015	1,356	1,634	1,826	2,038	2,255	2,520	3,484	4,379	5,610	4,443	7,753	15,994	25,022
Tax	-181	-147	-148	-196	-237	-265	-296	-327	-365	-505	-635	-814	-673	-1,124	-2,319	-3,628
Net Profit	961	782	867	1,160	1,397	1,561	1,743	1,928	2,154	2,979	3,744	4,797	3,770	6,629	13,675	21,394
EPS (NT\$)	7.14	5.81	6.44	8.61	10.38	11.59	12.94	14.32	16.00	22.13	27.81	35.63	28.01	49.23	101.56	158.89
Margins (%)																
Gross Margin	28.1%	33.3%	24.2%	19.0%	24.6%	23.8%	22.5%	22.4%	22.4%	24.0%	23.6%	25.3%	24.8%	23.2%	24.0%	23.6%
Operating Margin	15.6%	15.5%	12.2%	10.1%	13.4%	13.5%	13.2%	13.1%	13.4%	15.6%	15.5%	17.6%	12.7%	13.3%	15.8%	15.4%
Net Margin	13.7%	12.8%	10.1%	9.4%	12.2%	12.0%	11.7%	11.6%	11.8%	13.6%	13.4%	15.2%	11.0%	11.8%	13.7%	13.3%
Sequential Growth (%)																
Revenue	17%	-13%	41%	44%	-8%	14%	14%	12%	10%	20%	27%	13%	36%	64%	78%	61%
Gross Profit	-1%	3%	3%	13%	20%	10%	8%	11%	10%	28%	25%	21%	4%	54%	84%	59%
EBIT	25%	-14%	12%	19%	22%	15%	12%	11%	12%	39%	26%	29%	14%	71%	111%	57%
Net Profit	14%	-19%	11%	34%	20%	12%	12%	11%	12%	38%	26%	28%	9%	76%	106%	56%
EPS	14%	-19%	11%	34%	20%	12%	12%	11%	12%	38%	26%	28%	9%	76%	106%	56%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Alchip – N3 AI accelerator starts contributing in 2H26

We expect Alchip's revenue to snap back substantially in 2H26 thanks to its client's N3 AI accelerator ramp-up. Although its major CSP's N3 accelerator demand may reach 2m/4m units in 2026/2027, the capacity allocation will be critical for Alchip's growth. We also expect GM to be better than its previous product cycle for Alchip thanks to the company's back-end design contribution for more complex chip design.

Alchip – Maintain Buy and keep our TP unchanged at NT\$6,200. We raise our 2026/2027 EPS forecast by 14%/9% as we expect a solid ramp on of its client's N3 AI accelerator projects through 2H26 to 2027 but slightly lower our 2028E EPS on potential model transition. We maintain our Buy rating and keep our TP unchanged at NT\$6,200, based on our 37x PER to our 2027 EPS estimates (largely in line with its 3-year PER average).

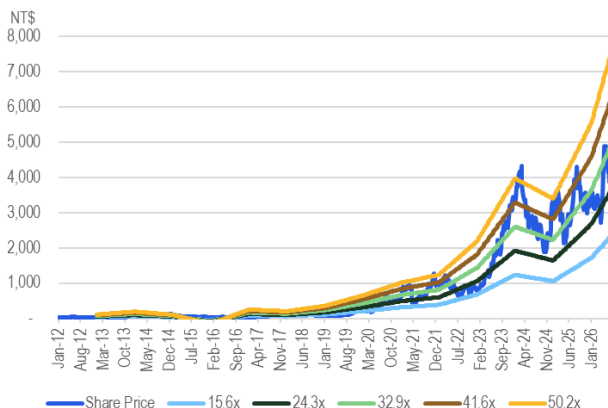
Figure 10. Alchip – Estimates Revisions

(NT\$m)	2026E			2027E			2028E		
	New	Old	Chg.	New	Old	Chg.	New	Old	Chg.
Sales	35,401	37,178	-5%	105,120	105,283	0%	155,878	155,495	0%
YoY growth	14%	20%		197%	183%		48%	48%	
Gross profit	12,030	10,882	11%	23,507	22,108	6%	33,042	33,819	-2%
Opex	2,937	3,063	-4%	7,582	7,595	0%	11,244	11,217	0%
Operating profit	9,094	7,819	16%	15,925	14,513	10%	21,798	22,603	-4%
Pre-tax profit	10,686	9,411	14%	17,593	16,181	9%	23,466	24,271	-3%
Net income	8,533	7,505	14%	13,969	12,845	9%	18,597	19,234	-3%
EPS (NT\$)	103.06	90.64	14%	168.72	155.14	9%	224.61	232.30	-3%
Gross margin	34.0%	29.3%	+4.7 ppt	22.4%	21.0%	+1.4 ppt	21.2%	21.7%	-0.6 ppt
Opex ratio	8.3%	8.2%	+0.1 ppt	7.2%	7.2%	-0.0 ppt	7.2%	7.2%	-0.0 ppt
Operating margin	25.7%	21.0%	+4.7 ppt	15.1%	13.8%	+1.4 ppt	14.0%	14.5%	-0.6 ppt
Net margin	24.1%	20.2%	+3.9 ppt	13.3%	12.2%	+1.1 ppt	11.9%	12.4%	-0.4 ppt

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

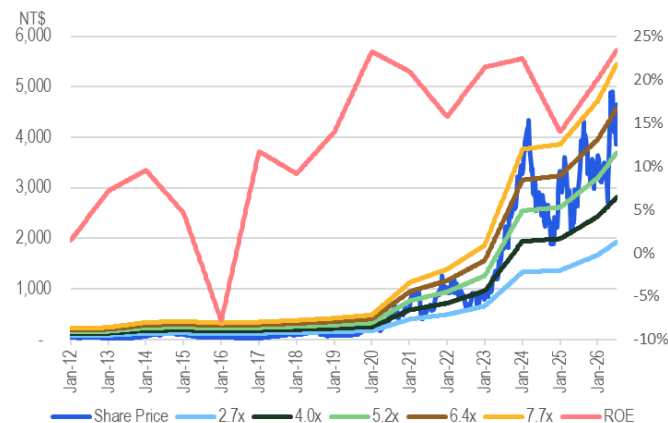
Figure 11. Alchip – Forward P/E band



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 12. Alchip – Forward P/B band & ROE



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Figure 13. Alchip – Forecast Summary

Alchip	2025				2026				2027				2025	2026E	2027E	2028E
(NT\$ in Mn, year-end Dec)	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q				
Revenue	10,485	9,144	6,570	4,727	4,187	6,029	9,421	15,764	20,069	25,190	28,420	31,441	30,926	35,401	105,120	155,878
COGS	-8,056	-7,257	-4,729	-2,730	-2,086	-3,417	-6,469	-11,399	-15,681	-19,890	-21,994	-24,049	-22,772	-23,371	-81,613	-122,836
Depreciation costs	-789	-543	-578	-667	-400	-500	-600	-700	-900	-900	-900	-900	-2,577	-2,200	-3,600	-3,600
Gross Profit	2,429	1,887	1,841	1,997	2,101	2,613	2,952	4,365	4,389	5,300	6,426	7,392	8,154	12,030	23,507	33,042
Operating Expense	-934	-737	-684	-782	-733	-422	-678	-1,103	-1,465	-1,713	-2,046	-2,358	-3,138	-2,937	-7,582	-11,244
SG&A expenses	-249	-251	-233	-197	-251	-181	-254	-394	-602	-630	-767	-943	-930	-1,081	-2,942	-4,371
R&D expenses	-467	-494	-453	-417	-482	-241	-424	-709	-863	-1,083	-1,279	-1,415	-1,831	-1,856	-4,640	-6,873
EBIT	1,494	1,150	1,157	1,215	1,368	2,191	2,274	3,261	2,924	3,587	4,380	5,034	5,017	9,094	15,925	21,798
Pre-Tax Profit	1,793	1,599	1,591	1,737	1,784	2,608	2,691	3,603	3,340	4,004	4,797	5,452	6,720	10,686	17,593	23,466
Tax	-332	-277	-265	-250	-357	-564	-570	-651	-741	-866	-1,015	-985	-1,124	-2,142	-3,607	-4,852
Net Profit	1,464	1,324	1,328	1,483	1,428	2,040	2,117	2,948	2,595	3,134	3,778	4,462	5,598	8,533	13,969	18,597
EPS - diluted (NT\$)	17.69	16.02	16.04	17.92	17.25	24.64	25.57	35.61	31.34	37.86	45.62	53.90	67.68	103.06	168.72	224.61
Margins (%)																
Gross Margin	23.2%	20.6%	28.0%	42.3%	50.2%	43.3%	31.3%	27.7%	21.9%	21.0%	22.6%	23.5%	26.4%	34.0%	22.4%	21.2%
Operating Margin	14.3%	12.6%	17.6%	25.7%	32.7%	36.3%	24.1%	20.7%	14.6%	14.2%	15.4%	16.0%	16.2%	25.7%	15.1%	14.0%
Net Margin	14.0%	14.5%	20.2%	31.4%	34.1%	33.8%	22.5%	18.7%	12.9%	12.4%	13.3%	14.2%	18.1%	24.1%	13.3%	11.9%
Sequential Growth (%)																
Revenue	-20%	-13%	-28%	-28%	-11%	44%	56%	67%	27%	26%	13%	11%	-40%	14%	197%	48%
Gross Profit	-12%	-22%	-2%	9%	5%	24%	13%	48%	1%	21%	21%	15%	-20%	48%	95%	41%
EBIT	-15%	-23%	1%	5%	13%	60%	4%	43%	-10%	23%	22%	15%	-23%	81%	75%	37%
Net Profit	-20%	-10%	0%	12%	-4%	43%	4%	39%	-12%	21%	21%	18%	-13%	52%	64%	33%
EPS	-24%	-9%	0%	12%	-4%	43%	4%	39%	-12%	21%	21%	18%	-14%	52%	64%	33%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, company data

Adding Upside 90-Day Short-Term View on Alchip (3661.TW)

Direction: Upside
Duration: Within 90 Days
Category: Thematic driven

We expect Alchip to see better revenues momentum as its client's N3 AI accelerator starts to ramp up from 2H26 to support its share price performance.

Short-Term View on GUC (3443.TW)

Direction: Upside
Duration: Within 90 Days (expires 26 Jul 2026)
Date Added: 26 Apr 2026
Category: Thematic driven

We are seeing strong growth momentum from GUC's CPU project with leading CSP client in the recent quarters to provide upside to its share price.

Catalyst Watch on Mediatek (2454.TW)

Direction: Upside
Duration: Within 90 Days (expires 30 Jul 2026)
Date Added: 30 Apr 2026
Catalyst: Thematic driven

We expect Mediatek's upbeat outlook on its AI ASIC business and its growth opportunity in CPO development should provide an upside catalyst for its share price.

Bull/Bear: Alchip (3661.TW)

NT\$ **7,000.00**
▲ 66% Upside

NT\$ **6,200.00**
▲ 47% Upside

NT\$ **4,000.00**
▼ 5.1% Downside



Spread 71pp
Current Price and expected returns (upside/downside) as of 06 Jul 2026

BULL Assumptions

- More-than-expected AI ASIC projects revenue contribution
- New projects wins from exist or new customers

BASE Assumptions

- Sales 2025-28E CAGR of around +50%

BEAR Assumptions

- Slower-than-expected NRE projects and revenue contribution
- Capacity constraint at advanced packaging
- More competition from its peers or/and insourcing at their customers

Bull/Bear: GUC (3443.TW)

NT\$ 6,200.00



▲ 35% Upside

NT\$ 6,000.00



▲ 30% Upside

NT\$ 3,500.00



▼ 24% Downside



Spread 59pp
Current Price and expected returns (upside/downside) as of 06 Jul 2026

BULL Assumptions



- Stronger projects demand from US CSP customers
- Stronger-than-expected NRE demand from HPC

BASE Assumptions



- Smooth project progress with US customers in advanced node demand

BEAR Assumptions



- Slower projects demand from US CSP customers
- Weaker-than-expected NRE demand from HPC

Bull/Bear: Mediatek (2454.TW)

NT\$ **6,500.00**
▲ 58% Upside

NT\$ **6,055.00**
▲ 47% Upside

NT\$ **4,000.00**
▼ 3.0% Downside



Spread 61pp
Current Price and expected returns (upside/downside) as of 06 Jul 2026

BULL Assumptions

- Higher than expected AI ASIC development
- Stronger than expected margins expansion
- Bull case TP is based on its peak cycle valuation of 37x PER to 2027/28E EPS

BASE Assumptions

- 2027/28E GM of >45%
- 2027/28E OPM of >16%

BEAR Assumptions

- Lower than AI ASIC development
- Worse than expected margins declines
- Bear case TP is based on 23x PER (the mid-point of its historical valuation) to 2027/28E EPS

Alchip

Company description

Alchip is a leading provider of silicon design and manufacturing service, specializing in ASIC (Application-Specific Integrated Circuit) design, development, and production. Founded in 2003, the company has >1,000 employees and operates in Taipei, Shanghai, Japan, Korea, and the US. Alchip serves a wide range of industries, including automotive, industrials, consumer electronics, and computing.

Investment strategy

We rate Alchip as Buy. We expect Alchip to deliver a strong snapback in 2026/27 with its N3 project ramping up. Moreover, we are also positive over the long-term growth opportunity from its AI ASIC projects with its major CSP client. Alchip remains one of the few design service providers within TSMC's 3DFabric alliance. In our view, if CSPs work more aggressively on their internal design and become more competitive on product offerings, Alchip would be well positioned as a back-end/turnkey partner.

Valuation

Our target price for Alchip of NT\$6,200 is based on 37x 2027E EPS. 37x target PER is largely in line with its historical 3-year PER range, which we see it as justified on account of strong growth potential of its AI ASIC projects with its key CSP client, with better project cycle and strong growth momentum resuming next year. Our target price equates to 2026/27E P/B of 10x/8x.

Risks

Key downside risks to our target price include: 1) slower-than-expected NRE projects and revenue contributions; 2) capacity constraints at advanced packaging; and 3) more competition and/or insourcing by customers.

GUC

Company description

Founded in 1998, Global Unichip (GUC) specializes in providing IC design services, particularly in ASIC service. As a subsidiary of TSMC, which holds 34.8% of GUC, it offers a wide range of design services including spec-in and SoC integration, physical implementation, 2.5D/3D packaging technologies, mass production services, and advanced technologies such as high bandwidth memory (HBM) and chip to chip interconnect IP. With its diversified global presence – Taiwan, North America, China, Europe, South Korea, and Japan – the company offers services including mass production, NRE, customer wafer verification projects, and IP licensing. Mass production services provide customers with complete service, from back-end design completion to wafer manufacturing, testing and management.

Investment strategy

We rate GUC stock as Buy/High Risk. We are seeing strong growth from its CPU projects with two leading CSP customers. We also see strong order

potential from its EV clients. Long-term, we still believe back-end design capability and 3DIC system integration will be critical to AI accelerators, which would give better opportunities for TSMC 3DFabric partners with GUC also being a beneficiary, in our view.

Valuation

Our target price for GUC of NT\$6,000 is based on 45x PER to the average of our 2027/28E EPS, slightly above its historical 3-year PER average which we believe is justified by its strong earnings outlook supported by solid demand from its AI projects with two US leading CSPs and growth upside from its US EV clients. Our target price equates to 2027/28E PB of 24x/15x.

Risks

Our quant risk rating system rates the GUC stock High Risk for its share-price volatility. Downside risks that could mean the stock trades below our target price include: 1) slower projects demand from US CSP customers, 2) lower-than-expected NRE demand from less AI ASIC projects, and 3) delayed or less turnkey revenues contribution from key customers.

Mediatek

Company description

MediaTek (MTK) is a fabless IC company that designs SoC for smartphone, smart TV, and digital media devices. It also develops chips for connectivity, networking, smart home gadgets, and automotive sectors. It was founded in 1997, is listed on the Taiwan Stock Exchange, and is headquartered in Taiwan. It has sales and research subsidiaries in mainland China, Singapore, India, the US, Japan, Korea, Denmark, and England. It primarily sells its products in China and other emerging-market countries.

Investment strategy

We rate MediaTek shares as Buy. We expect Mediatek to ride on 5G migration in emerging markets and market share gain of its high-end Dimensity series in China market as well. In particular, we are seeing strong growth potential from its AI ASIC, NTN-satellite business in the long term. We believe solid fundamentals and strong operating leverage will support Mediatek's margins outlook.

Valuation

Our NT\$6,055 target price for MediaTek is based on 35x P/E to 2027/2028 EPS, which is 1.5-std above its five-year average forward P/E given the promising outlook from its AI ASIC business starting from 2H26. We believe our target PE is also justified by Mediatek's global and regional peers' trading at an average of 36x PER. AI ASIC development with CSPs should provide more upside from 2027 onwards. We also see Mediatek well prepared for 6G and 6G+NTN Satellite communication. Despite lukewarm smartphone overall demand, we expect Mediatek to benefit from 5G migration in emerging markets and its Dimensity series should also gain shares in flagship models. Non-smartphone business is also seeing stabilizing TV and IoT demand with

Mediatek outgrowing the overall market on its better technology in WiFi 7 and connectivity.

Risks

Key downside risks that could prevent the shares from reaching our target price include: 1) weaker-than-expected growth from AI ASIC business; 2) more severe competition in the smartphone segment; 3) slower overall demand related to macroeconomic risks, especially for smartphone demand; and 4) further price hikes from foundries.

Appendix A-1

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by “AC” in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst’s compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

Analyst: Laura (Chia Yi) Chen



	Date	Rating	Target Price	Closing Price
1	28-Jul-23 15:06:57	*3H	*1,540.00	1,815.00
2	10-Dec-23 22:11:06	3H	*1,440.00	1,895.00
3	25-Apr-24 19:13:09	*1H	*1,500.00	1,200.00
4	20-Jun-24 04:12:46	1H	*2,040.00	1,625.00

	Date	Rating	Target Price	Closing Price
5	26-Jul-24 10:14:16	1H	*1,760.00	1,335.00
6	07-Feb-25 13:35:43	1H	*1,750.00	1,380.00
7	24-Apr-25 18:56:50	1H	*1,200.00	964.00
8	02-Jul-25 10:54:37	*3H	*1,225.00	1,325.00

	Date	Rating	Target Price	Closing Price
9	07-Oct-25 23:46:05	*1H	*1,850.00	1,385.00
10	02-Jan-26 11:31:34	1H	*2,725.00	2,185.00
11	26-Apr-26 21:10:44	1H	*5,050.00	4,035.00
12	01-May-26 14:30:23	1H	*5,550.00	4,260.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

Analyst: Laura (Chia Yi) Chen



	Date	Rating	Target Price	Closing Price
1	20-Aug-23 18:42:38	1	*2,310.00	1,970.00
2	25-Sep-23 23:40:21	1	*3,080.00	2,625.00
3	02-Nov-23 19:35:26	1	*3,280.00	2,820.00
4	10-Dec-23 22:11:06	1	*3,840.00	3,335.00
5	20-Feb-24 08:20:35	1	*4,850.00	4,150.00
6	25-Apr-24 19:13:09	1	*3,900.00	2,830.00

	Date	Rating	Target Price	Closing Price
7	25-Aug-24 17:11:26	1	*3,700.00	2,655.00
8	24-Oct-24 11:30:35	*2	*2,000.00	1,980.00
9	01-Nov-24 09:43:49	2	*2,200.00	2,030.00
10	09-Feb-25 15:55:03	2	*3,815.00	3,605.00
11	27-Feb-25 10:20:56	2	*3,700.00	3,285.00
12	01-May-25 14:24:22	2	*2,400.00	2,125.00

	Date	Rating	Target Price	Closing Price
13	02-Jul-25 10:54:37	2	*3,180.00	3,190.00
14	13-Aug-25 10:10:55	*1	*4,800.00	4,120.00
15	06-Mar-26 09:46:22	1	*4,200.00	3,310.00
16	08-May-26 12:27:59	1	*6,200.00	4,890.00

*Indicates Change

Rating/target price changes above reflect Eastern Time

Mediatek (2454.TW)

Ratings and Target Price History Fundamental Research

Analyst: Laura (Chia Yi) Chen



	Date	Rating	Target Price	Closing Price
1	28-Jul-23 11:22:19	2	*660.00	658.00
2	20-Sep-23 14:57:20	2	*735.00	759.00
3	24-Oct-23 12:02:27	*1	*935.00	816.00
4	06-Nov-23 16:50:42	1	*1,030.00	882.00
5	10-Mar-24 16:15:26	1	*1,420.00	1,230.00
6	11-Sep-24 10:08:27	1	*1,250.00	1,110.00

*Indicates Change

	Date	Rating	Target Price	Closing Price
7	30-Oct-24 10:35:19	1	*1,420.00	1,290.00
8	02-Jan-25 08:42:43	1	*1,580.00	1,350.00
9	07-Feb-25 11:57:14	1	*1,800.00	1,525.00
10	30-Apr-25 10:15:39	1	*1,600.00	1,350.00
11	02-Jul-25 10:54:37	1	*1,450.00	1,295.00
12	30-Jul-25 08:56:32	1	*1,540.00	1,380.00

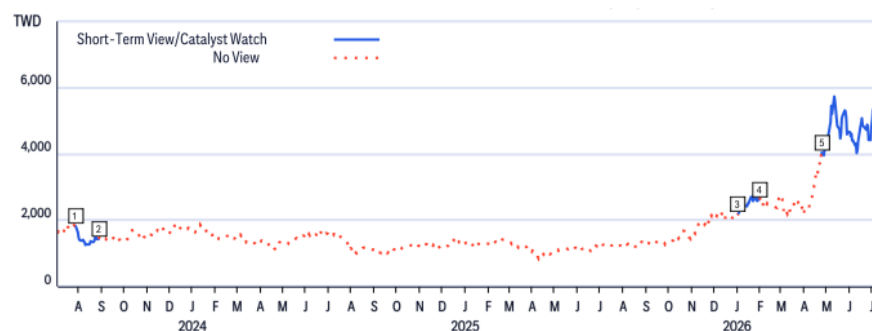
	Date	Rating	Target Price	Closing Price
13	28-Jan-26 07:35:51	1	*2,200.00	1,780.00
14	04-Feb-26 07:24:16	1	*2,350.00	1,800.00
15	01-Apr-26 13:20:42	1	*2,125.00	1,465.00
16	26-Apr-26 21:10:44	1	*2,800.00	2,435.00
17	30-Apr-26 11:24:47	1	*3,100.00	2,610.00
18	27-May-26 09:26:19	1	*5,950.00	4,640.00

Rating/target price changes above reflect Eastern Time

GUC (3443.TW)

Short-Term View/Catalyst Watch Research

Analyst: Laura (Chia Yi) Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	29-Jul-23 01:08:51	Add CW	Downside	30 Days	1,815.00
2	28-Aug-23 11:44:27	Remove CW	Downside	30 Days	1,430.00

CW - Catalyst Watch, STV - Short-Term View

	Date	Action	Expected Direction	Duration	Closing Price
3	02-Jan-26 06:31:34	Add STV	Upside	30 Days	2,185.00
4	01-Feb-26 19:57:44	Remove STV	Upside	30 Days	2,620.00

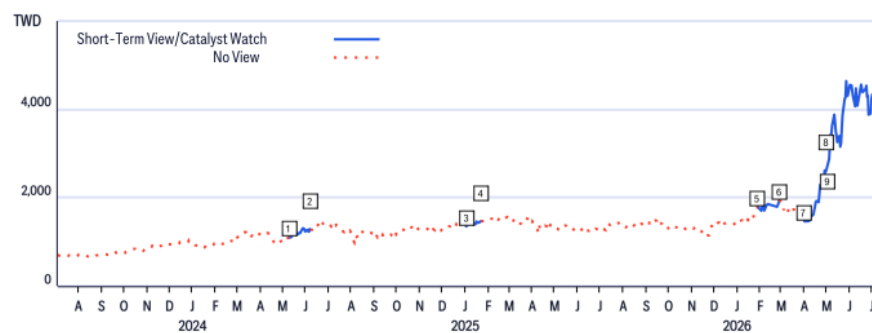
	Date	Action	Expected Direction	Duration	Closing Price
5	26-Apr-26 17:10:44	Add STV	Upside	90 Days	4,035.00

Rating/target price changes above reflect Eastern Time

Mediatek (2454.TW)

Short-Term View/Catalyst Watch Research

Analyst: Laura (Chia Yi) Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	10-May-24 03:08:23	Add CW	Upside	30 Days	1,095.00
2	07-Jun-24 12:03:57	Remove CW	Upside	30 Days	1,275.00
3	02-Jan-25 03:42:43	Add CW	Upside	30 Days	1,350.00

CW - Catalyst Watch, STV - Short-Term View

	Date	Action	Expected Direction	Duration	Closing Price
4	31-Jan-25 12:14:36	Remove CW	Upside	30 Days	1,465.00
5	28-Jan-26 02:35:51	Add STV	Upside	30 Days	1,780.00
6	27-Feb-26 12:27:51	Remove STV	Upside	30 Days	1,945.00

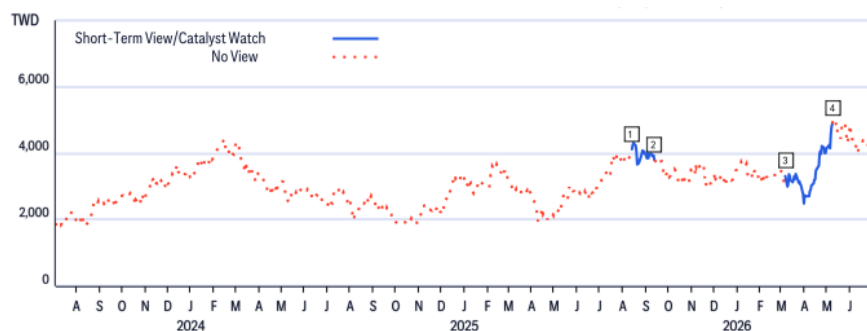
	Date	Action	Expected Direction	Duration	Closing Price
7	01-Apr-26 09:20:42	Add STV	Upside	90 Days	1,465.00
8	30-Apr-26 07:24:47	Remove STV	Upside	90 Days	2,610.00
9	30-Apr-26 07:24:47	Add CW	Upside	90 Days	2,610.00

Rating/target price changes above reflect Eastern Time

Alchip (3661.TW)

Short-Term View/Catalyst Watch Research

Analyst: Laura (Chia Yi) Chen



	Date	Action	Expected Direction	Duration	Closing Price
1	13-Aug-25 06:10:55	Add STV	Upside	30 Days	4,120.00
2	12-Sep-25 14:04:30	Remove STV	Upside	30 Days	3,790.00
3	06-Mar-26 04:46:22	Add STV	Downside	90 Days	3,310.00
4	08-May-26 08:27:59	Remove STV	Downside	90 Days	4,890.00

CW - Catalyst Watch, STV - Short-Term View

Rating/target price changes above reflect Eastern Time

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from Alchip, GUC, Mediatek in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: GUC, Mediatek.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: Alchip, GUC, Mediatek.

Citigroup Global Markets Inc. and/or its affiliates has a significant financial interest in relation to Mediatek. (For an explanation of the determination of significant financial interest, please refer to the policy for managing conflicts of interest which can be found at www.citiVelocity.com.)

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (<https://www.citivelocity.com/cv2>) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

Data current as of 01 Jul 2026	12 Month Rating			Catalyst Watch		
	Buy	Hold	Sell	Buy	Hold	Sell
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	7%	36%	48%	16%
% of companies in each rating category that are investment banking clients	38%	42%	27%	42%	37%	36%

Guide to Citi Research Fundamental Research Investment Ratings:

Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.

Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.

Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus

the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign "Rating Suspended" status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as "—" and "—" respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned "Under Review" status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views ("STV") Ratings Disclosure:

Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be different from and does not affect a stock's fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Taiwan Securities Company Limited Jack Chen; Laura (Chia Yi) Chen; Nicholas Lai

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy", "Hold", "Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q1 2026 Buy 33%(63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33% (61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%); Q2 2025 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.4% (86%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy"

means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (<https://www.citivelocity.com/>). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968 and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without

notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA - Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar(parte) - CEP: 01311920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia : This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Superintendencia Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to "securities" (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section "Research Analyst Affiliations / Non-US Research Analyst Disclosures" for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM's registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai - 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures, controls, and training to ensure continued compliance with all applicable rules and regulations. All recommendations contained herein are made by duly qualified research analysts. CGM's Corporate Identity Number is U99999MH2000PTC126657, and its Compliance Officer [Vishal Bohra] contact details are:

Tel: +91-022-61759994, Fax: +91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at <https://www.citivelocity.com/rendition/authfilelinksvcs/eppublic/V1/file?paramData=ZmlsZU5hbWU9L1MzL0dETVMvcHVjRGZlY2xvc3VyZUh0bWxzL2dkbV9kaXNfc2ViaV9UZjtc29mVXNlMh0bWw>

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of

CGMK. <http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/DISFundMgrAnalystList.xml&divisionId=MDISO3002002000000&serviceld=SDISO3002002000>. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comisión Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul. Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold/Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana, Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E.**, these materials (the "Materials") are communicated by Citigroup Global Markets Limited, DIFC branch ("CGML"), an entity registered in the Dubai International Financial Center ("DIFC") and licensed and regulated by the Dubai Financial Services Authority ("DFSA", license #CLO221) to Professional Clients and Market Counterparties, as defined in DFSA regulations, only and should not be relied upon or distributed to Retail Clients. Financial products and/or services to which the Materials relate will only be made available to Professional Clients and Market Counterparties. Citigroup Global Markets Limited DIFC Branch registered address is Level 3, Gate District Building 02, Dubai International Financial Centre and can be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested. The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt, investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information

confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information <(e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.

Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, disseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST